

Exercises 2B:

1) Find all the vector spaces that have exactly one basis.

Let V be a vector space such that

$$v_1, \dots, v_m \in V \quad a_1 v_1 + \dots + a_m v_m = v \quad \forall a_1, \dots, a_m \in \mathbb{R}$$
$$\forall v \in V, \quad a_1, \dots, a_m \text{ is unique}$$

$$\therefore \text{if } a_1 v_1 + \dots + a_m v_m = 0$$

$$\text{then } a_1, \dots, a_m = 0$$

①

for $v_1 + v_2, v_2, \dots, v_m$

$$a_1 v_1 + a_2 v_2 + a_2 v_2 + \dots + a_m v_m = 0$$

then

$$a_1 v_1 + (a_1 + a_2) v_2 + \dots + a_m v_m = 0$$

this is linearly independent as it is equivalent to ①

from
rows
swap
exchange.

and if $v_1, \dots, v_m$ spans the vector space so does $v_1 + v_2, v_2, \dots, v_m$

So for $m > 1$ there is more than one basis

but $m=1$ then the list is v_1 where $v_1 \neq 0$
otherwise $a_1 v_1 = 0$ is non-unique.

vectors spans of the form

$$V = \{0, v_1\}$$

then have the basis of v_1 and all other vectors in V are of the form $a \cdot v_1$

2) Verify all the assertions in

2.27 example: bases

- (a) The list $(1, 0, \dots, 0), (0, 1, 0, \dots, 0), \dots, (0, \dots, 0, 1)$ is a basis of F^n , called the *standard basis* of F^n .
- (b) The list $(1, 2), (3, 5)$ is a basis of F^2 . Note that this list has length two, which is the same as the length of the standard basis of F^2 . In the next section, we will see that this is not a coincidence.
- (c) The list $(1, 2, -4), (7, -5, 6)$ is linearly independent in F^3 but is not a basis of F^3 because it does not span F^3 .
- (d) The list $(1, 2), (3, 5), (4, 13)$ spans F^2 but is not a basis of F^2 because it is not linearly independent.
- (e) The list $(1, 1, 0), (0, 0, 1)$ is a basis of $\{(x, x, y) \in F^3 : x, y \in F\}$.
- (f) The list $(1, -1, 0), (1, 0, -1)$ is a basis of $\{(x, y, z) \in F^3 : x + y + z = 0\}$.
- (g) The list $1, z, \dots, z^m$ is a basis of $\mathcal{P}_m(F)$, called the *standard basis* of $\mathcal{P}_m(F)$.

$$a) (1, 0, \dots, 0), (0, 1, 0, \dots, 0), \dots, (0, \dots, 0, 1)$$

is linearly independent as

$$a_1(1, 0, \dots, 0) + a_2(0, 1, 0, \dots, 0) + \dots + a_n(0, \dots, 0, 1)$$

$$= (a_1, a_2, \dots, a_n) \text{ therefore for } (a_1, a_2, \dots, a_n) = 0$$

$$a_1 = 0, a_2 = 0, \dots, a_n = 0$$

$(a_1, a_2, \dots, a_n)$ Clearly spans F^n as let $(\omega_1, \omega_2, \dots, \omega_n) \in F^n, \forall \omega_1, \omega_2, \dots, \omega_n \in F$

$$\text{then set } a_1 = \omega_1, a_2 = \omega_2, \dots, a_n = \omega_n$$

$$b) (1,2), (3,5) \in F^2$$

Linear independence

$$\text{for } a_1, a_2 \in F \text{ let } a_1(1,2) + a_2(3,5) = 0$$

$$\text{then } (a_1, 2a_1) + (3a_2, 5a_2) = 0$$

$$\therefore a_1 + 3a_2 = 0, \quad 2a_1 + 5a_2 = 0 \quad \therefore a_1 = -3a_2$$

$$a_1 = 0 = a_2$$

Span
for $(w_1, w_2) \in F^2$

$$\text{let } a_1(1,2) + a_2(3,5) = (w_1, w_2)$$

$$\therefore a_1 + 3a_2 = w_1, \quad 2a_1 + 5a_2 = w_2$$

$$a_1 = w_1 - 3a_2$$

$$a_2 = \frac{w_2 - 2a_1}{5}$$

$$a_2 = \frac{w_2 - 2w_1 + 6a_2}{5}$$

$$a_2 = 2w_1 - w_2$$

$$a_1 = w_1 - 6w_1 + w_2$$

$$w_1 = \frac{a_2 + w_2}{2}$$

$$a_1 = w_2 - 5w_1$$

$$a_1 = w_2 - \frac{5}{2}(a_2 + w_2)$$

$$w_2 = -\frac{7}{3}a_1$$

$$\frac{7}{2}a_1 = -\frac{3}{2}w_2$$

$\therefore (1,2), (3,5)$ spans F^2 and is a basis.

$$c) (1, 2, -4), (7, -5, 6) \in F^3$$

linear independence

$$a_1(1, 2, -4) + a_2(7, -5, 6) = 0$$

$$a_1 + 7a_2 = 0 \quad \therefore a_1 = -7a_2$$

$$2a_1 - 5a_2 = 0 \quad -14a_2 - 5a_2 = 0, a_2 = 0$$

$$-4a_1 + 6a_2 = 0 \quad \therefore -4a_1 = 0, a_1 = 0$$

Span

$$\text{Let } (\omega_1, \omega_2, \omega_3) \in F^3$$

$$\text{for } (\omega_1, \omega_2, \omega_3) = a_1(1, 2, -4) + a_2(7, -5, 6)$$

$$\text{Let } \omega_1 = 1, \omega_2 = 1, \omega_3 = 0$$

$$\text{then } \textcircled{I} a_1 + 7a_2 = 1 \quad \textcircled{III} \therefore 4a_1 = 6a_2$$

$$\textcircled{II} 2a_1 - 5a_2 = 1$$

$$a_2 = \frac{2}{3}a_1$$

$$\textcircled{III} -4a_1 + 6a_2 = 0 \quad \textcircled{I} \therefore \frac{14}{3}a_1 + a_1 = 1$$

$$\therefore a_2 = \frac{2}{17}$$

$$= a_1 = \frac{3}{17}$$

$$\textcircled{II} \text{ but } 2\left(\frac{3}{17}\right) - 5\left(\frac{2}{17}\right) = \frac{-4}{17} \neq 1$$

So it does not span F^3

$$d) (1, 2), (3, 5), (4, 13) \in F^2$$

Span

$$\text{let } (\omega_1, \omega_2) \in F^2$$

$$a_1(1, 2) + a_2(3, 5) + a_3(4, 13) = (\omega_1, \omega_2)$$

$$\forall a_1, a_2, a_3, a_4 \in F$$

$$\therefore a_1 + 3a_2 + 4a_3 = \omega_1 \quad \text{let } a_3 = -\frac{3}{4}a_2$$

$$2a_1 + 5a_2 + 13a_3 = \omega_2 \quad \text{then}$$

$$\text{and } 2a_1 + 5a_2 - 13\left(\frac{3}{4}a_2\right) = \omega_2 \quad a_1 = \omega_1$$

$$2a_1 - \frac{19}{4}a_2 = \omega_2 = 2\omega_1 - \frac{19}{4}a_2$$

therefore it spans F^2

$$\text{by letting } a_1 = \omega_1 \quad \text{and } a_2 = (2\omega_1 - \omega_2) \frac{4}{19}$$

$$\text{and } a_3 = -\frac{3}{4}a_2$$

Linear Independence

$$\text{let } a_1(1, 2) + a_2(3, 5) + (4, 13) = 0$$

$$\text{for } a_1, a_2 \in F$$

$$\therefore a_1 + 3a_2 + 4 = 0 \quad \therefore a_1 = -3a_2 - 4$$

$$2a_1 + 5a_2 + 13 = 0 \quad \therefore -a_2 + 5 = 0 \quad a_2 = 5 \neq 0$$

$$\therefore a_1 = -15 + 4 = -11 \neq 0 \quad \text{therefore not linearly independent.}$$

e) $(1, 1, 0), (0, 0, 1) \in F^3$

Let $U = \{(x, x, y) \in F^3 \mid x, y \in F\}$

Linear independence

for $a_1, a_2 \in F$

$$\begin{aligned} \text{If } a_1(1, 1, 0) + a_2(0, 0, 1) &= 0 \\ &= (a_1, a_1, a_2) \quad \therefore a_1 = a_2 = 0 \end{aligned}$$

Span $\therefore$ linearly independent.

$$a_1(1, 1, 0) + a_2(0, 0, 1) = (x, x, y)$$

$$\therefore a_1 = x \quad a_1 = x, a_2 = y \quad \forall x, y \in F$$

$$\therefore \text{Spans } U$$

f) $(1, -1, 0), (1, 0, -1), U = \{(x, y, z) \in F^3 \mid x + y + z = 0\}$

Linear independence

Let $a_1, a_2 \in F$

$$\text{If } a_1(1, -1, 0) + a_2(1, 0, -1) = 0$$

$$\therefore (a_1, -a_1, 0) + (a_2, 0, -a_2) = 0$$

$$\therefore (a_1 + a_2, -a_1, -a_2) = 0$$

$$a_1 + a_2 = 0 \therefore a_1 = -a_2$$

$$-a_1 = 0 \therefore a_1 = 0 \therefore a_2 = 0$$

$\therefore$ linearly independent.

Span

$$a_1(1, -1, 0) + a_2(1, 0, -1) = (x, y, z)$$

$$\text{Such that } x + y + z = 0$$

$$\therefore x = -y - z$$

$$\therefore (a_1 + a_2, -a_1, -a_2) = (x, y, z)$$

$$a_1 + a_2 = -(-a_1) - (-a_2)$$

$\therefore$ setting $a_1 = -y$ and $a_2 = -z$
the last spans U

$$g) 1, z, \dots, z^m \in \mathcal{P}_m(F)$$

Linear independence

$$\text{For } a_0, a_1, \dots, a_m \in F$$

$$\text{If } a_0 + a_1 z + \dots + a_m z^m = 0$$

Setting $a_0 + a_1 z + \dots + a_m z^m = v^m = a_0, a_1, \dots, a_m$

$$\therefore a_0, a_1, \dots, a_m = (0, \dots, 0)$$

then

$$a_0 = 0, a_1 = 0, \dots, a_m = 0$$

3) a) Let U be the subspace of $\mathbb{R}^5$ defined by

$$U = \{(x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5 \mid x_1 = 3x_2 \text{ and } x_3 = 7x_4\}$$

Find a basis for U

Let $v_1 = (3, 1, 0, 0, 0)$ $v_2 = (0, 0, 0, 0, 1)$

$$v_3 = (0, 0, 7, 1, 0), v_1, v_2, v_3 \in U$$

$$\therefore (3, 1, 0, 0, 0), (0, 0, 7, 1, 0), (0, 0, 0, 0, 1) = v_1, v_2, v_3$$

and for $a_1, a_2, a_3 \in \mathbb{F}$ is linearly independent.

$$a_1(3, 1, 0, 0, 0)$$

$$+ a_2(0, 0, 7, 1, 0) = (3a_1, a_1, 7a_2, a_2, a_3)$$

$$+ a_3(0, 0, 0, 0, 1)$$

$\therefore$ spans U and is a basis of U

b) Extend the basis in (a) to a basis of $\mathbb{R}^5$

$$\text{Let } v_4 = (1, 0, 0, 0, 0), v_5 = (0, 0, 1, 0, 0)$$

$$v_1, v_2, v_3, v_4, v_5 = (3, 1, 0, 0, 0), (0, 0, 7, 1, 0), (0, 0, 0, 0, 1), (1, 0, 0, 0, 0), (0, 0, 1, 0, 0)$$

c) Find a subspace W of $\mathbb{R}^5$ such that $\mathbb{R}^5 = U \oplus W$.

$$W = \left\{ (x, 0, y, 0, 0) \in \mathbb{R}^5 \mid x, y \in \mathbb{R} \right\}$$

Prove that $\mathbb{R}^5 = U \oplus W$

First

$$\begin{aligned} U + W &= \left\{ (x_1, x_2, x_3, x_4, x_5) \mid x_1, \dots, x_5 \in \mathbb{R} \right\} \\ &= \mathbb{R}^5 \end{aligned}$$

then that $U \cap W = \{0\}$

$$\text{when } x = 3(0) \quad x = 0 \quad \therefore U \cap W$$

$$\begin{aligned} \text{when } y = 7(0) \quad y = 0 \\ U \cap W &= (0, 0, 0, 0, 0) \\ &\parallel \\ &\{0\} \end{aligned}$$

4) a) Let U be the subspace of $\mathbb{C}^5$ defined

$$U = \{(z_1, z_2, z_3, z_4, z_5) \in \mathbb{C}^5 \mid 6z_1 = z_2 \text{ and } z_3 + 2z_4 + 3z_5 = 0\}$$

Find a basis of U

$$\text{Let } v_1 = (1, 6, 0, 0, 0), v_2 = (0, 0, -12, 3, 2)$$

both are ^{obviously} linearly independent.

Span

for $a+bi, x+yi \in \mathbb{C}$

$$(a+bi)(1, 6, 0, 0, 0) + (x+yi)(0, 0, -12, 3, 2)$$

$$= (a+bi, 6a+6bi, 0, 0, 0) + (0, 0, -12x-12yi, 3x+3yi, 2x+2yi)$$

$$= (a+bi, 6a+6bi, -12x-12yi, 3x+3yi, 2x+2yi)$$

$$= (z_1, z_2, z_3, z_4, z_5)$$

$$6z_1 = 6a+6bi = z_2, \quad z_3 + 2z_4 + 3z_5 = -12x-12yi + 6x+6yi + 6x+6yi = 0$$

$$\therefore (1, 6, 0, 0, 0), (0, 0, -12, 3, 2) = 0$$

Spans U

b) extend the basis in (a) to a basis of $\mathbb{C}'^5$

$$\mathbb{C}'^5 = \{z_1, z_2, z_3, z_4, z_5 \mid z_1, \dots, z_5 \in \mathbb{C}'\}$$

$$\text{Let } v_3 = (0, 1, 0, 0, 0), v_4 = (0, 0, 1, 0, 0) \\ v_5 = (0, 0, 0, 0, 1)$$

c) Find a subspace W of $\mathbb{C}'^5$ such that
 $\mathbb{C}'^5 = U \oplus W$

$$W = \{(x, 0, y, 0, 0) \in \mathbb{C}'^5 \mid x, y \in \mathbb{C}'\}$$

$$U + W = \{(a, b, c, d, e) \in \mathbb{C}'^5 \mid a, b, c, d, e \in \mathbb{C}'\}$$

$$\forall (x, 0, y, 0, 0) \quad 6x = 0 \quad \text{only when} \\ \text{and} \quad y + 2x0 + 3x0 = 0 \quad x = 0$$

$$\text{when } y = 0$$

$$\therefore U \cap W = \{0\}$$

$$\therefore U \oplus W = \mathbb{C}'^5$$

5) Suppose V is finite-dimensional and U, W are subspaces of V , such that $V = U + W$.

Prove that there exists a basis of V consisting of vectors in $U \cup W$.

Since $V = U + W$ and a basis v must be in the span of V then it must be in U or W so is also in $U \cup W$.

is this incomplete?

6) Prove or give a counterexample. If p_0, p_1, p_2, p_3 is a list in $\mathcal{P}_3(F)$ such that none of the polynomials has degree 2, then p_0, p_1, p_2, p_3 is not a basis of $\mathcal{P}_3(F)$.

$$\begin{aligned} \text{Let } p_0 &= 1 & \text{then } \text{Span}(p_0, p_1, p_2, p_3) \\ p_1 &= z & = a_0 + a_1 z + a_2 z^2 + (a_2 + a_3) z^3 \\ p_2 &= z^2 + z^3 & \text{which is equivalent to } \mathcal{P}_3(F) \\ p_3 &= z^3 \end{aligned}$$

$$\mathcal{P}_3(F) = \{c_0 + c_1 z + c_2 z^2 + c_3 z^3 \mid c_0, c_1, c_2, c_3 \in F, \forall z \in F\}$$

The list is also linearly independent since

$$\text{If } a_0 + a_1 z + a_2 z^2 + (a_2 + a_3) z^3 = 0 \text{ then } a_0 = 0, a_1 = 0, a_2 = 0, \therefore a_3 = 0$$

7) Suppose v_1, v_2, v_3, v_4 is a basis of V .

Prove that $v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$
is also a basis of V .

If v_1, v_2, v_3, v_4 is a basis of V

then if $a_1 v_1 + a_2 v_2 + a_3 v_3 + a_4 v_4 = 0$

$$a_1, a_2, a_3, a_4 = 0$$

①

② and

$$\forall a_1, a_2, a_3, a_4 \in F \quad \text{Span}(v_1, v_2, v_3, v_4) = V$$

for $v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$

$$\text{Span} = a_1 v_1 + a_1 v_2 + a_2 v_2 + a_2 v_3 + a_3 v_3 + a_3 v_4 + a_4 v_4$$

$$= a_1 v_1 + (a_1 + a_2) v_2 + (a_2 + a_3) v_3 + (a_3 + a_4) v_4$$

Let

$$c_1 = a_1$$

$$\therefore = c_1 v_1 + c_2 v_2 + c_3 v_3 + c_4 v_4$$

$$c_2 = a_1 + a_2$$

which is equivalent to ②

$$c_3 = a_2 + a_3$$

$$\forall c_1, c_2, c_3, c_4 \in F$$

$$c_4 = a_3 + a_4$$

then for $a_1 v_1 + a_1 v_2 + a_2 v_2 + a_2 v_3 + a_3 v_3 + a_3 v_4 + a_4 v_4 = 0$

$$a_1 = 0 \quad \therefore a_2 = 0 \quad \therefore a_3 = 0 \quad \therefore a_4 = 0$$

8) Prove or give a counterexample. If v_1, v_2, v_3, v_4 is a basis of V , and U is a subspace of V such that $v_1, v_2 \in U$ and $v_3, v_4 \notin U$ then v_1, v_2 is a basis of U .

from online solution (I thought I had proved it with a 2 page mistake lol)

$$\text{Let } V = \mathbb{R}^4, \quad v_1 = (1, 0, 0, 0) \\ v_2 = (0, 1, 0, 0) \\ v_3 = (0, 0, 1, 0) \\ v_4 = (0, 0, 0, 1)$$

$$\text{Let } U = \left\{ (x_1, x_2, x_3, 2x_3) \in \mathbb{R}^4 \mid x_1, x_2, x_3 \in \mathbb{R} \right\} \quad (\text{is a subspace})$$

$$v_1, v_2 \in U$$

$$v_3, v_4 \notin U$$

$$\forall a_1, a_2 \in \mathbb{R} \quad a_1 v_1 + a_2 v_2 = \left\{ (a_1, a_2, 0, 0) \in \mathbb{R}^4 \mid a_1, a_2 \in \mathbb{R} \right\} = X$$

$$X \neq U \quad \text{Since } (1, 1, 1, 2) \in U, \notin X$$

$$\text{So } \text{Span}(v_1, v_2) \neq U$$

9) Suppose $v_1, \dots, v_m$ is a list of vectors in V . For $k \in \{1, \dots, m\}$
let $w_k = v_1 + \dots + v_k$.

Show that $v_1, \dots, v_m$ is a basis of $V \iff w_1, \dots, w_m$ is a basis of V .

Linear independence

Let $v_1, \dots, v_m$ be linearly independent

then if $a_1 v_1 + \dots + a_m v_m = 0$ 1, $a_1, \dots, a_m \in F$
 $a_1, \dots, a_m = 0$

for $w_k = v_1 + \dots + v_m$, $\forall k \in \{1, \dots, m\}$

then for $c_1, \dots, c_m \in F$

$$c_1 w_1 + \dots + c_m w_m = c_1 v_1 + c_2 v_1 + c_2 v_2 + \dots + c_m v_1 + \dots + c_m v_m$$

$$\therefore = (c_1 + \dots + c_m) v_1 + (c_2 + \dots + c_m) v_2 + \dots + c_m v_m$$

$$\text{letting } c_1 = a_1 - (c_2 + \dots + c_m)$$

$$c_2 = a_2 - (c_3 + \dots + c_m)$$

$\vdots$

$$c_m = a_m, \quad \forall a_1, \dots, a_m \in F$$

$$\therefore = a_1 v_1 + \dots + a_m v_m \quad (\text{equivalent to } \textcircled{1})$$

$\therefore v_1, \dots, v_m$ is linearly independent $\iff w_1, \dots, w_m$ is linearly independent.

Span

then similarly If $a_1 v_1 + \dots + a_n v_n = v \quad \forall v \in V$
 $\forall a_1, \dots, a_n \in F$

then $\forall a_1, \dots, a_n$ let $c_1 = a_1 - (c_2 + \dots + c_n)$
 $\vdots$
 $c_n = a_n$

then

$$c_1 \omega_1 + \dots + c_n \omega_n = v, \quad \forall v \in V, \forall c_1, \dots, c_n \in F$$

$$\therefore \text{Span}(v_1, \dots, v_n) = V \iff \text{Span}(\omega_1, \dots, \omega_n) = V$$

10) Suppose U and W are subspaces of V such that $V = U \oplus W$. Suppose also that $u_1, \dots, u_m$ is a basis of U and $\omega_1, \dots, \omega_n$ is a basis of W .

Prove that $u_1, \dots, u_m, \omega_1, \dots, \omega_n$ is a basis of V .

If $V = U \oplus W$ then $U + W = V$ and $U \cap W = \{0\}$

If $\text{Span}(u_1, \dots, u_m) = U$ and $\text{Span}(\omega_1, \dots, \omega_n) = W$

then $\exists a_1, \dots, a_m \in F, \exists b_1, \dots, b_n \in F$
 $a_1 u_1 + \dots + a_m u_m = u \quad b_1 \omega_1 + \dots + b_n \omega_n = \omega \quad \forall u \in U, \forall \omega \in W$

Since $V = U + W$ then $u + \omega = v \quad \forall v \in V, \exists u \in U, \omega \in W$

$$\therefore v = a_1 u_1 + \dots + a_m u_m + b_1 \omega_1 + \dots + b_n \omega_n, \quad \forall v \in V$$

$$\therefore \text{Span}(u_1, \dots, u_n, w_1, \dots, w_m) = V$$

Since $U \cap W = \{0\}$ then $\forall w_1, \dots, w_m \neq 0, w_1, \dots, w_m \notin U$
and $\forall u_1, \dots, u_n \neq 0, u_1, \dots, u_n \notin W$

therefore since both are closed under addition
then

$$w_k \neq a_1 u_1 + \dots + a_n u_n \quad \forall a_1, \dots, a_n \in F, k \in \{1, \dots, m\}$$

and

$$u_k \neq b_1 w_1 + \dots + b_m w_m \quad \forall b_1, \dots, b_m \in F \quad (\text{assuming both} \neq 0)$$

So

$u_1, \dots, u_n, w_1, \dots, w_m$ is linearly independent.

$\therefore u_1, \dots, u_n, w_1, \dots, w_m$ is a basis of V .

11) Suppose V is a real vector space. Show that if $v_1, \dots, v_n$ is a basis of V (as a real vector space), then $v_1, \dots, v_n$ is also a basis of the complexification $V_{\mathbb{C}}$ (as a complex vector space)

Since $v_1, \dots, v_n$ is a basis of V (as a real vector space)

then $\forall c_1, \dots, c_n \in \mathbb{R}$

$$c_1 v_1 + \dots + c_n v_n = v, \quad \forall v \in V \quad (I)$$

Where for $v=0$ $c_1, \dots, c_n = 0$ (unique) (II)

now for $z_1, \dots, z_n \in \mathbb{C}$ s.t. $a_1, b_1, \dots, a_n, b_n \in \mathbb{R}$

$$z_1 v_1 + \dots + z_n v_n = a_1 v_1 + b_1 i v_1 + \dots + a_n v_n + b_n i v_n = v_{\mathbb{C}} \quad (III)$$

Span

$$\forall v_{\mathbb{C}} \in V_{\mathbb{C}} \quad v_{\mathbb{C}} = \operatorname{Re}(v_{\mathbb{C}}) + \operatorname{Im}(v_{\mathbb{C}}) = v + i \operatorname{Im}(v_{\mathbb{C}})$$

$$= a_1 v_1 + \dots + a_n v_n + \operatorname{Im}(v_{\mathbb{C}}) i, \quad \forall a_1, \dots, a_n \in \mathbb{R}$$

The imaginary part of $v_{\mathbb{C}}$ is just every possible vector in V multiplied by i

$$\therefore \operatorname{Im}(v_{\mathbb{C}}) = (c_1 v_1 + \dots + c_n v_n) i \quad \forall c_1, \dots, c_n \in \mathbb{R}$$

$$= c_1 v_1 i + \dots + c_n v_n i \quad \text{letting } b_1 = c_1, \dots, c_n = b_n$$

then $V_C = a_1 v_1 + \dots + a_m v_m + b_1 v_i + \dots + b_n v_n$ (equivalent to) III

$$\therefore \text{Span}(v_1, \dots, v_m) = V_C$$

linear independence

$$\text{Let } z_1 v_1 + z_2 v_2 + \dots + z_m v_m = 0$$

$$= a_1 v_1 + \dots + a_m v_m + b_1 v_i + \dots + b_n v_n = 0$$

$$= a_1 v_1 + \dots + a_m v_m + (b_1 v_i + \dots + b_n v_n) = 0 + 0$$

from II

$$\therefore = \underbrace{[a_1 v_1 + \dots + a_m v_m]}_{a_1, \dots, a_m = 0} + \underbrace{[b_1 v_i + \dots + b_n v_n]}_{b_1, \dots, b_n = 0} = 0 + 0$$

$$\therefore \text{When } z_1 v_1 + \dots + z_m v_m = 0 \text{ then } z_1, \dots, z_m = 0$$

$\therefore$ linear independent in V_C

$\therefore v_1, \dots, v_m$ is a basis of V_C if $v_1, \dots, v_m$ is a basis of V .